

# CoRoT-2b

R-1068

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_CoRoT-2\_151020 · created 2026-07-30T13:06:43+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457315.6706 +/- 0.0019 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.143 +/- 0.065                 |
| Transit depth (Rp/Rs)^2                           | 2.04 +/- 1.85 [%]               |
| Orbital Inclination (inc)                         | 83.46 +/- 2.08                  |
| Airmass coefficient 1 (a1)                        | 1.33 +/- 0.73                   |
| Airmass coefficient 2 (a2)                        | -0.123 +/- 0.087                |
| Scatter in the residuals of the lightcurve fit is | 56.84 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 3.0                             |
| Optimal Annulus                                   | 5.75                            |
| Transit Duration (day)                            | 0.064 +/- 0.025                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.143 ± 0.065 vs 0.1667 (deviation 0.34σ) |
| Duration fit vs published | 0.064 d vs 0.095 d (deviation 1.14σ)      |
| Mid-time O–C              | 2.1 min                                   |
| Depth SNR                 | 2.0                                       |
| Residual scatter          | 56.84 %                                   |

### Light curve

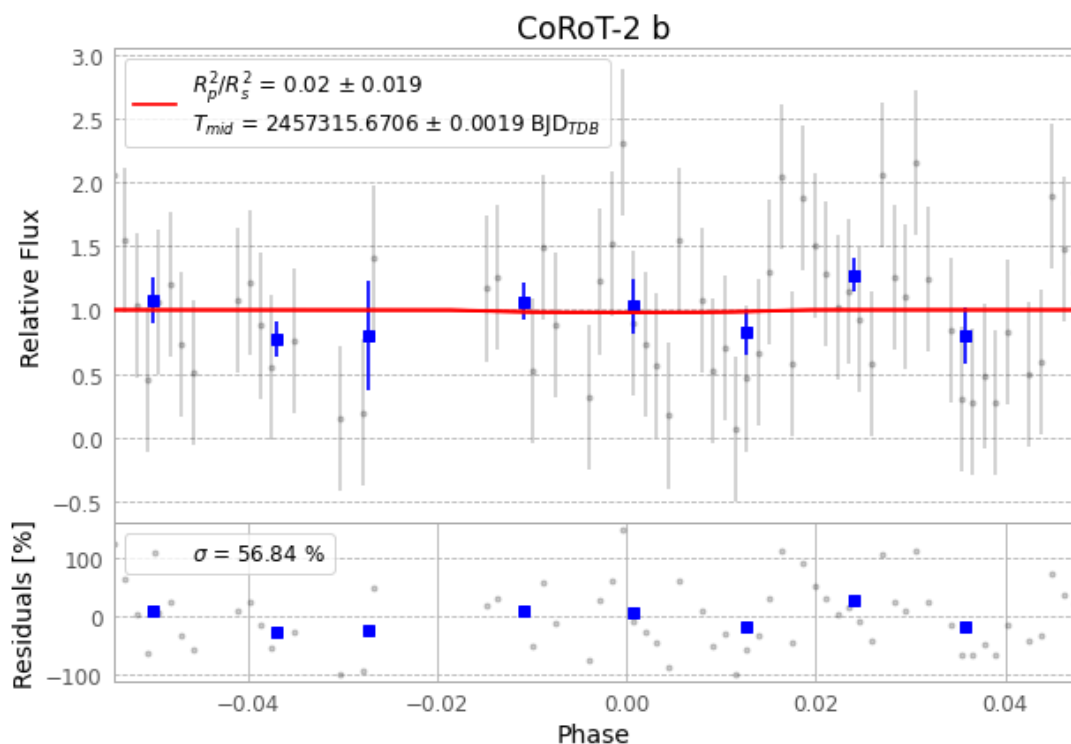


Figure 1 — FinalLightCurve\_CoRoT-2 b\_2015-10-19.png (SHA-256 575526313e80f4ad...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 458], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 d6ce392179cc9247...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

86 FITS frames (56 MB), CoRoT-2151020014814.FITS → CoRoT-2151020060313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-151020.log (fc3b5eb78a51...), FinalLightCurve\_CoRoT-2 b\_2015-10-19.png (575526313e80...), FinalLightCurve\_CoRoT-2 b\_2015-10-19.csv (f8f887c40eba...)

## Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 13:06Z.